

```
In [108]: import pandas as pd
import numpy as np
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
import string
import re
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.feature_extraction.text import CountVecorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
import matplotlib.pyplot as plt
from wordcloud import WordCloud
nltk.download('punkt')
nltk.download('stopwords')

[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\massi\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]   C:\Users\massi\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!

Out[108]: True
```

Importiamo il dataset

```
In [2]: data_set = pd.read_csv("C:/Users/massi/OneDrive/Desktop/Classificare i reclami dei consumatori con python e il Machine Learning/Lamentele_sample.csv")
```

Visualizziamo le prime osservazioni del data_set

0	2686509	2021-10-04	Credit reporting, credit repair services, or o...	Credit reporting	Improper use of your report	Reporting company used your report improperly	NaN
1	2107155	2017-01-19	Student loan	Federal student loan servicing	Dealing with my lender or servicer	Received bad information about my loan	In 2009 I got married and changed my last name...
2	1116803	2018-01-31	Credit reporting, credit repair services, or o...	Credit reporting	Unable to get your credit report or credit score	Problem getting your free annual credit report	NaN
3	299024	2022-03-16	Credit reporting, credit repair services, or o...	Credit reporting	Problem with a credit reporting company's inve...	Their investigation did not fix an error on yo...	NaN
4	4251	2022-11-03	Credit reporting, credit repair services, or o...	Credit reporting	Improper use of your report	Reporting company used your report improperly	NaN

Dimensioni del data_set -> Numero di righe x Numero di colonne

Dimensioni del data_set -> Numero di righe x Numero di colonne

```
In [7]: data_set.shape
Out[7]: (10000, 7)
```

Visualizziamo la struttura del data_set

```
In [8]: data_set.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 7 columns):
 #   Column                Non-Null Count  Dtype
---  --
 0   Unnamed: 0            10000 non-null  int64
 1   Date received         10000 non-null  object
 2   Product               10000 non-null  object
 3   Sub-product           9248 non-null   object
 4   Issue                 10000 non-null  object
 5   Sub-issue             7797 non-null   object
 6   Consumer complaint narrative  3619 non-null   object
dtypes: int64(1), object(6)
memory usage: 547.0+ KB
```

Notiamo la presenza di valori Mancanti (NA)

Rimuoviamo tali valori dal data_set

```
In [9]: data_set_clean = data_set.dropna()
```

```
In [10]: data_set_clean.shape
Out[10]: (2905, 7)
```

Le features di interesse sono: Product e Consumer complaint narrative

```
In [18]: data_set_product_narrative = data_set_clean[['Product', 'Consumer complaint narrative']].reset_index(drop=True)
```

```
In [19]: data_set_product_narrative
```

	Product	Consumer complaint narrative
0	Student loan	In 2009 I got married and changed my last name...
1	Debt collection	This agency is calling repeated times during t...
2	Credit reporting, credit repair services, or o...	The account for XXXX XXXX XXXX XXXX, in which...
3	Credit reporting, credit repair services, or o...	Department of Education XXXX XXXX XXXX XXXX X...
4	Credit reporting, credit repair services, or o...	On XXXX XXXX XXXX XXXX I gave permission for a cred...
...	...	...
2900	Credit reporting, credit repair services, or o...	I am a victim of identity theft. I have been p...
2901	Checking or savings account	I have regions bank- life green checking acco...
2902	Credit reporting, credit repair services, or o...	15usc1681b no permission to place on consumer ...
2903	Credit reporting, credit repair services, or o...	I have been requesting information from the bu...
2904	Credit reporting, credit repair services, or o...	!Hello im writing to you because my student lo...

2905 rows x 2 columns

In questo caso, per ogni prodotto abbiamo la narrazione dei reclami dei consumatori

Per un dato Prodotto, quante narrazioni di reclami abbiamo ?

```
In [26]: data_set_product_narrative.groupby('Product').size().sort_values(ascending=False)
```

```
Out[26]: Product
Credit reporting, credit repair services, or other personal consumer reports    1670
Debt collection                                                                235
Credit card or prepaid card                                                    175
Checking or savings account                                                     99
Student loan                                                                    92
Vehicle loan or lease                                                           82
Payday loan, title loan, or personal loan                                      3
dtype: int64
```

La maggioranza delle narrazioni dei reclami dei consumatori riguardano: Credit reporting, credit repair services, or other personal consumer reports, Debt Collection, Credit card or prepaid card

Prendiamo in considerazione i primi 3 Prodotti che hanno ricevuto il maggior numero di reclami

```
In [116]: product_list = ['Credit reporting, credit repair services, or other personal consumer reports', 'Debt collection', 'Credit card or prepaid card']
```

```
In [118]: filtered_df = data_set_product_narrative[data_set_product_narrative['Product'].isin(product_list)].copy()
```

Preprocessing del testo

```
In [78]: nltk.download('punkt')
nltk.download('stopwords')
stemmer = nltk.stem.porter.PorterStemmer("english")
stop_words = set(stopwords.words('english'))

[nltk_data] Downloading package punkt to
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[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
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[nltk_data]   Package stopwords is already up-to-date!

In [142]: def clean(text):
text = str(text).lower() # Convertire il contenuto text in stringa (nel caso non lo fosse) e poi lo trasforma tutto in minuscolo.
text = re.sub(r'[^\w\s]','', text, flags=re.IGNORECASE) # Rimuove X,X
text = re.sub(r'\(|\)|\?|\\', '', text) # Rimuove tutto ciò che è racchiuso tra parentesi quadre
text = re.sub(r'http(s)?://[^\w\.\s]+', '', text) # Rimuove tutti i link che iniziano con "http://" o "https://" o "www."
text = re.sub(r'<.*?>+', '', text) # Rimuove tag HTML o simili racchiusi tra <...>
text = re.sub(r'(\s|\\|) re.escape(string.punctuation)', '', text) # Rimuove tutti i segni di punteggiatura
text = re.sub(r'(\n|,|',',', text) # Elimina i caratteri di a capo (\n)
text = re.sub(r'Verida', '', text) # Rimuove tutte le parole che contengono almeno una cifra (0-9).
text = [word for word in text.split(' ') if word not in stop_words] # Divide il testo in parole (split(' ')) e crea una lista con solo le parole che non sono stopwords.
text = " ".join(text) # Word cloud in textem: la lista di parole filtrate in una singola stringa, separando con spazi.
text = (stemmer.stem(word) for word in text.split(' ')) # Riduce una parola alla sua radice o forma base
text = " ".join(text)
return text

In [143]: filtered_df['Consumer complaint narrative'] = filtered_df['Consumer complaint narrative'].apply(clean)
```

```
In [144]: filtered_df
```

	Product	Consumer complaint narrative
1	Debt collection	agenc call repeat time day multipl attempt cea...
2	Credit reporting, credit repair services, or o...	account im author user report correct credit ...
3	Credit reporting, credit repair services, or o...	depart educ:pa multitud problem stu...
4	Credit reporting, credit repair services, or o...	gave permis credit bureau check credit home ...
5	Credit reporting, credit repair services, or o...	howev account inquiri compani dont believ acco...
...	...	...
2899	Credit reporting, credit repair services, or o...	recent obtain credit report nolic duplic accou...
2900	Credit reporting, credit repair services, or o...	victim ident theft pull polic share name someo...
2902	Credit reporting, credit repair services, or o...	permis place consum report privaci law broke...
2903	Credit reporting, credit repair services, or o...	request inform bureaus havent sent anyth sever...
2904	Credit reporting, credit repair services, or o...	!hello im write student loan payment report cr...

2546 rows x 2 columns

Conversione in vettori (Vectorization)

Trasformiamo il testo in numeri

Bag of Words (CountVectorizer) -> Primo Approccio

TF-IDF (TfidfVectorizer) -> Secondo Approccio

Bag of Words (CountVectorizer)

Si prende un corpus di testi, in questo caso i reclami per ogni prodotto e si costruisce un vocabolario con tutte le parole uniche.

Per ogni documento (singolo reclamo) si conta quante volte ogni parola del vocabolario costruito compare all'interno del documento in esame.

Quindi il numero di righe è pari al numero di documenti e per ogni documento abbiamo tot colonne (una per ogni vocabolo).

Se all'interno del documento i-esimo non compare un vocabolo specifico, avremo 0, viceversa 1/2/3... in base a quante volte quel vocabolo compare all'interno del documento.

```
In [145]: filtered_df = filtered_df[['Consumer complaint narrative', 'Product']]
x = np.array(filtered_df['Conaumer complaint narrative'])
y = np.array(filtered_df['Product'])
cv = CountVectorizer()
X = cv.fit_transform(x)
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.25, random_state=42)
```

Stima con il modello di regressione logistica (Iterazione massima = 1000)

```
In [155]: model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
Out[155]: LogisticRegression(max_iter=1000)
```

Previsione sul test set con report delle metriche di performance

```
In [156]: y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
Credit card or prepaid card	0.87	0.54	0.67	63
Credit reporting, credit repair services, or other personal consumer reports	0.85	0.95	0.90	411
Debt collection	0.79	0.68	0.73	163
accuracy				637
macro avg	0.84	0.72	0.77	637
weighted avg	0.84	0.84	0.83	637

Classificazione in base ad un dato testo in input

```
In [157]: new_text = "I requested a copy of my credit report from your agency, but it contains several inaccurate entries that are seriously affecting my credit score."
new_text_clean = clean(new_text)
new_vec = vectorizer.transform([new_text_clean])
pred = model.predict(new_vec)
print(pred)

['Credit reporting, credit repair services, or other personal consumer reports']

La previsione è corretta, il testo riguarda la categoria -> Credit reporting, credit repair services, or other personal consumer reports
```

```
In [158]: new_text = "I have been repeatedly contacted by your debt collection agency regarding a debt that I do not recognize."
new_text_clean = clean(new_text)
new_vec = vectorizer.transform([new_text_clean])
pred = model.predict(new_vec)
print(pred)

['Debt collection']

La previsione è corretta, il testo riguarda la categoria -> Debt collection
```

TF-IDF (TfidfVectorizer)

TF -> Conta quante volte una parola appare in un documento. Spesso normalizzato per la lunghezza del documento.

IDF -> Misura quanto una parola è rara o comune nel corpus.

Se una parola appare molto spesso in un documento ma è rara nel resto del corpus, allora avremo un alto peso TF-IDF, dunque importante per descrivere quel documento.

Se una parola appare in quasi tutti i documenti, allora avremo un basso peso TF-IDF, poco discriminante.

Trasforma ogni documento in un vettore numerico in cui ogni componente è il TF-IDF di una parola.

```
In [159]: vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(x)
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.25, random_state=42)
```

Stima con il modello di regressione logistica (Iterazione massima = 1000)

```
In [160]: model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
Out[160]: LogisticRegression(max_iter=1000)
```

Previsione sul test set con report delle metriche di performance

```
In [161]: y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
Credit card or prepaid card	0.87	0.54	0.67	63
Credit reporting, credit repair services, or other personal consumer reports	0.85	0.95	0.90	411
Debt collection	0.79	0.68	0.73	163
accuracy				637
macro avg	0.84	0.72	0.77	637
weighted avg	0.84	0.84	0.83	637

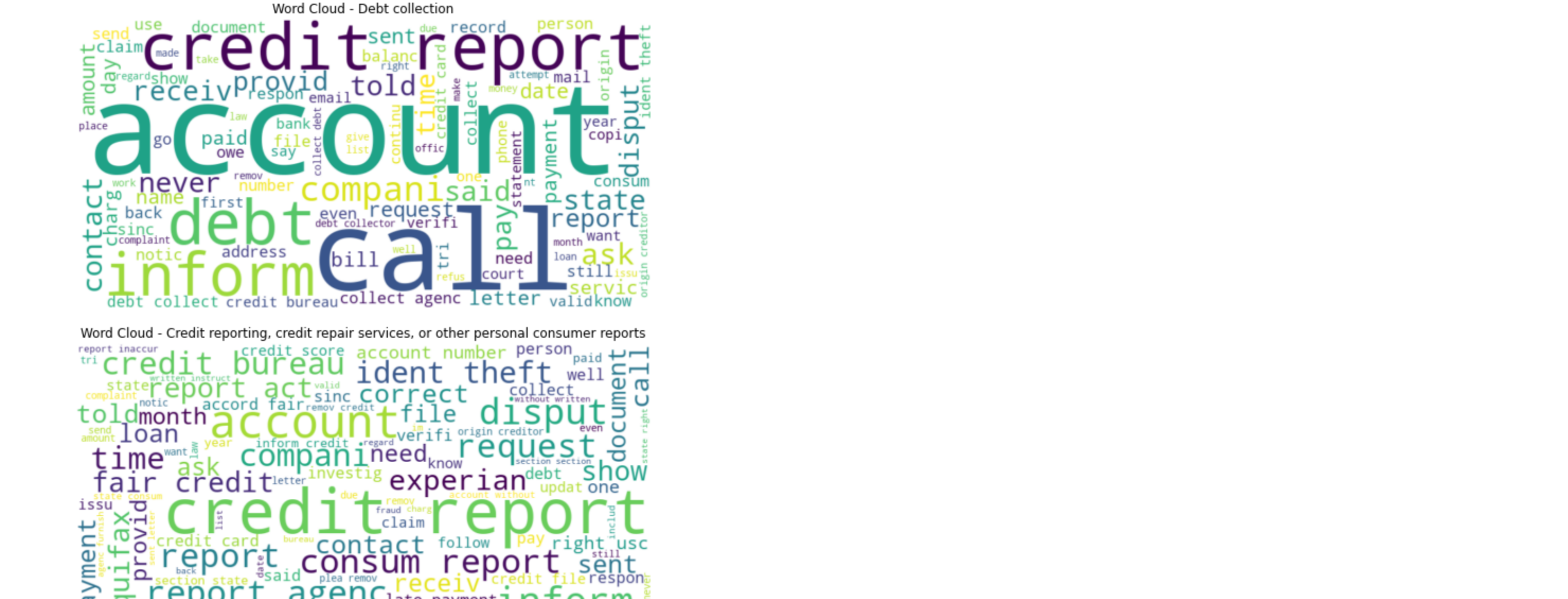
```
In [162]: new_text = "I have been repeatedly contacted by your debt collection agency regarding a debt that I do not recognize."
new_text_clean = clean(new_text)
new_vec = vectorizer.transform([new_text_clean])
pred = model.predict(new_vec)
print(pred)

['Debt collection']

WORLD CLOUD
```

Per ogni prodotto, costruiamo una world cloud.

Notiamo i vocaboli più utilizzati nei reclami in base al prodotto considerato.



```
In [ ]:
```